

EEE4114F: Problem set 6 solutions

Question 1

(a)

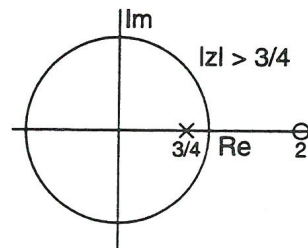
$$x[n] = \left(\frac{1}{2}\right)^n u[n] + (2)^n u[-n-1]$$

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{z}{z-2}, \quad \frac{1}{2} < |z| < 2$$

$$y[n] = 6 \left(\frac{1}{2}\right)^n u[n] - 6 \left(\frac{3}{4}\right)^n u[n]$$

$$Y(z) = \frac{6}{1 - \frac{1}{2}z^{-1}} - \frac{6}{1 - \frac{3}{4}z^{-1}}, \quad \frac{3}{4} < |z|$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{-\frac{3}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{3}{4}z^{-1})} \cdot \frac{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}{-\frac{3}{2}z^{-1}} = \frac{1 - 2z^{-1}}{1 - \frac{3}{4}z^{-1}}, \quad |z| > \frac{3}{4}$$



(b)

$$H(z) = \frac{1}{1 - \frac{3}{4}z^{-1}} - \frac{2z^{-1}}{1 - \frac{3}{4}z^{-1}}, \quad |z| > \frac{3}{4}$$

$$h[n] = \left(\frac{3}{4}\right)^n u[n] - 2 \left(\frac{3}{4}\right)^{n-1} u[n-1]$$

(c)

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - 2z^{-1}}{1 - \frac{3}{4}z^{-1}}$$

$$Y(z) - \frac{3}{4}z^{-1}Y(z) = X(z) - 2z^{-1}X(z)$$

$$y[n] - \frac{3}{4}y[n-1] = x[n] - 2x[n-1]$$

- (d) The ROC is outside $|z| = \frac{3}{4}$, which includes the unit circle. Therefore the system is stable. The $h[n]$ we found in part (b) tells us the system is also causal.

Question 2

$$y[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{4}\right)^n u[n] + u[n]$$

$$Y(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} + \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

$$x[n] = u[n]$$

$$X(z) = \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3 - \frac{19}{6}z^{-1} + \frac{2}{3}z^{-2}}{1 - \frac{7}{12}z^{-1} + \frac{1}{12}z^{-2}}, \quad |z| > \frac{1}{3}$$

- (a) Cross multiplying and equating z^{-1} with a delay in time:

$$y[n] - \frac{7}{12}y[n-1] + \frac{1}{12}y[n-2] = 3x[n] - \frac{19}{6}x[n-1] + \frac{2}{3}x[n-2]$$

- (b) Using partial fractions on $H(z)$ we get:

$$H(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} - \frac{z^{-1}}{1 - \frac{1}{3}z^{-1}} + \frac{1}{1 - \frac{1}{4}z^{-1}} - \frac{z^{-1}}{1 - \frac{1}{4}z^{-1}} + 1, \quad |z| > \frac{1}{3}$$

So,

$$h[n] = \left(\frac{1}{3}\right)^n u[n] - \left(\frac{1}{3}\right)^{n-1} u[n-1] + \left(\frac{1}{4}\right)^n u[n] - \left(\frac{1}{4}\right)^{n-1} u[n-1] + \delta[n]$$

- (c) Since the ROC of $H(z)$ includes $|z| = 1$ the system is stable.

Question 3

- (a) *It cannot be determined.* The ROC might or might not include the unit circle.
- (b) *It cannot be determined.* The ROC might or might not include $z = \infty$.
- (c) *False.* Given that the system is causal, we know that the ROC must be outside the outermost pole. Since the outermost pole is outside the unit circle, the ROC will not include the unit circle, and thus the system is not stable.
- (d) *True.* If the system is stable, the ROC must include the unit circle. Because there are poles both inside and outside the unit circle, any ROC including the unit circle must be a ring. A ring-shaped ROC means that we have a two-sided system.

Question 4

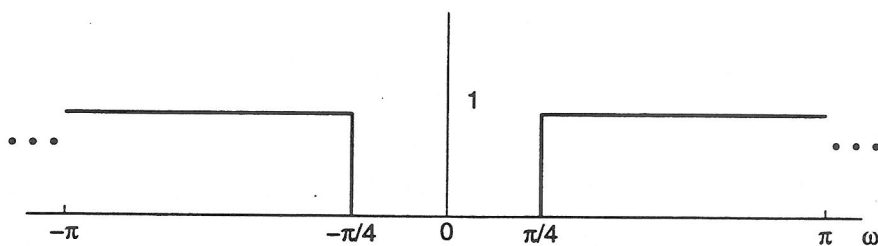
A minimum phase system is one which has all its poles and zeros inside the unit circle. It is causal, stable, and has a causal and stable inverse.

- (a) $H_1(z)$ has a zero outside the unit circle at $z = 2$ so it is not minimum phase.
- (b) $H_2(z)$ is minimum phase since its poles and zeros are inside the unit circle.
- (c) $H_3(z)$ is minimum phase since its poles and zeros are inside the unit circle.
- (d) $H_4(z)$ has a zero outside the unit circle at $z = \infty$ so it is not minimum phase. Moreover, the inverse system would not be causal due to the pole at infinity.

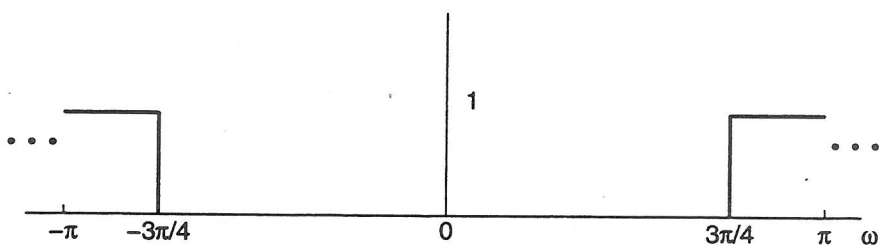
Question 5

$h_{lp}[n]$ is an ideal lowpass filter with $\omega_c = \frac{\pi}{4}$

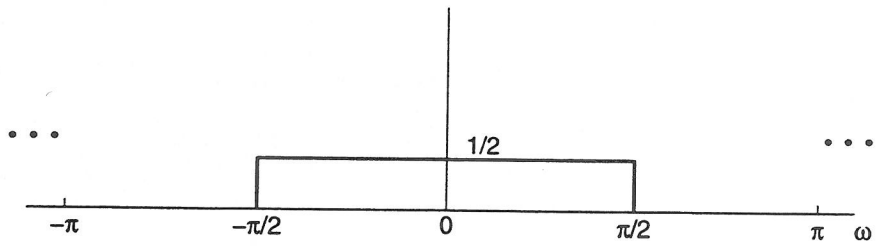
- (a) $y[n] = x[n] - x[n] * h_{lp}[n] \Rightarrow H(e^{j\omega}) = 1 - H_{lp}(e^{j\omega})$
This is a highpass filter.



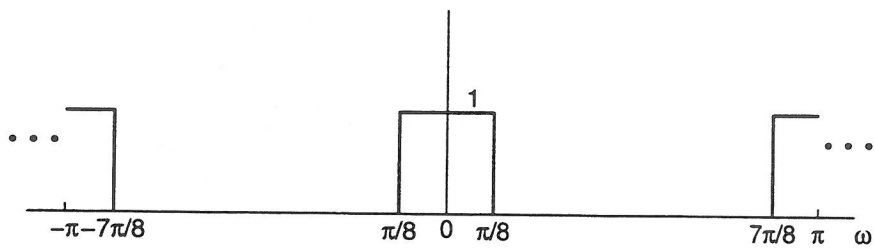
- (b) $x[n]$ is first modulated by π , lowpass filtered, and demodulated by π . Therefore, $H_{lp}(e^{j\omega})$ filters the high frequency components of $X(e^{j\omega})$.
This is a highpass filter.



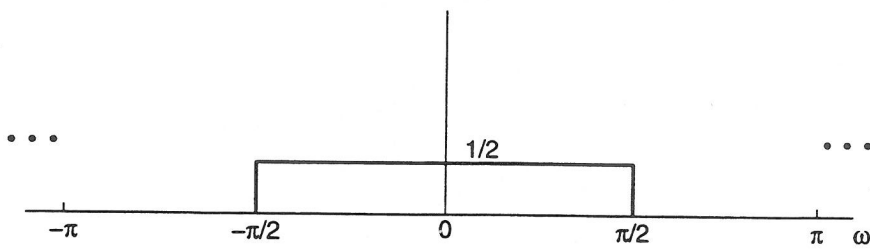
- (c) $h_{lp}[2n]$ is a downsampled version of the filter. Therefore, the frequency response will be "spread out" by a factor of two, with a gain of $\frac{1}{2}$.
This is a lowpass filter.



- (d) This system upsamples $h_{lp}[n]$ by a factor of two. Therefore, the frequency axis will be compressed by a factor of two. This is a bandstop filter.



- (e) This system upsamples the input before passing it through $h_{lp}[n]$. This effectively doubles the frequency bandwidth of $H_{lp}(e^{j\omega})$. This is a lowpass filter.



Question 6

(a)

$$H(z) = \frac{A}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{3}z^{-1})}, \quad |z| > \frac{1}{2} \quad h[n] \text{ causal}$$

$$H(1) = 6 \Rightarrow A = 4$$

(b)

$$H(z) = \frac{4}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{3}z^{-1})}, \quad |z| > \frac{1}{2}$$

$$= \frac{(\frac{12}{5})}{1 - \frac{1}{2}z^{-1}} + \frac{(\frac{8}{5})}{1 + \frac{1}{3}z^{-1}}$$

$$h[n] = \frac{12}{15} \left(\frac{1}{2}\right)^n u[n] + \frac{8}{5} \left(-\frac{1}{3}\right)^n u[n]$$

(c) (i)

$$x[n] = u[n] - \frac{1}{2}u[n-1] \Leftrightarrow X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 - z^{-1}}, \quad |z| > 1$$

$$Y(z) = X(z)H(z)$$

$$= \frac{1 - \frac{1}{2}z^{-1}}{1 - z^{-1}} \cdot \frac{4}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{3}z^{-1})}, \quad |z| > 1$$

$$= \frac{4}{(1 - z^{-1})(1 + \frac{1}{3}z^{-1})}$$

$$= \frac{3}{1 - z^{-1}} + \frac{1}{1 + \frac{1}{3}z^{-1}}$$

$$y[n] = 3u[n] + \left(-\frac{1}{3}\right)^n u[n]$$

(ii)

$$x(t) = 50 + 10 \cos(20\pi t) + 30 \cos(40\pi t)$$

$$T = \frac{1}{40} \quad t = nT$$

$$x[n] = 50 + 10 \cos \frac{\pi}{2}n + 30 \cos \pi n$$

$$= 50 + 5e^{j(n\pi/2)} + 5e^{-j(n\pi/2)} + 15e^{jn\pi} + 15e^{-jn\pi}$$

Using the eigenfunction property:

$$y[n] = 50H(e^{j0}) + 5e^{j(n\pi/2)}H(e^{j(\pi/2)}) + 5e^{-j(n\pi/2)}H(e^{-j(\pi/2)}) + 15e^{jn\pi}H(e^{j\pi}) + 15e^{-jn\pi}H(e^{-j\pi})$$

$$H(e^{j\omega}) = \frac{4}{1 - \frac{1}{6}e^{-j\omega} - \frac{1}{6}e^{-j2\omega}}$$

$$H(e^{j0}) = 6, H(e^{j(\pi/2)}) = 7\left(\frac{12}{25}\right) - j\frac{12}{25}, H(e^{-j(\pi/2)}) = 7\left(\frac{12}{25}\right) + j\frac{12}{25},$$

$$H(e^{j\pi}) = 4, H(e^{-j\pi}) = 4$$

$$y[n] = 300 + 24\sqrt{2} \cos\left(\frac{\pi}{2}n - \tan^{-1}\left(\frac{1}{7}\right)\right) + 120 \cos \pi n$$

Question 7

The type II FIR system $H_{II}(z)$ has generalized linear phase. Therefore, it can be written in the form

$$H_{II}(e^{j\omega}) = A_e(e^{j\omega})e^{-j\omega M/2}$$

where M is an odd integer and $A_e(e^{j\omega})$ is a real, even, periodic function of ω . Note that the system $G(z) = (1 - z^{-1})$ is a type IV generalized linear phase system.

$$\begin{aligned} G(e^{j\omega}) &= 1 - e^{-j\omega} \\ &= e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2}) \\ &= e^{-j\omega/2}(2j \sin(\omega/2)) \\ &= 2 \sin(\omega/2) e^{-j\omega/2 + j\pi/2} \\ &= A_o(e^{j\omega}) e^{-j\omega/2 + j\pi/2} \\ A_o(e^{j\omega}) &= 2 \sin(\omega/2) \\ \angle G(e^{j\omega}) &= -\frac{\omega}{2} + \frac{\pi}{2} \end{aligned}$$

The cascade of $H_{II}(z)$ with $G(z)$ results in a generalized linear phase system $H(z)$.

$$\begin{aligned} H(e^{j\omega}) &= A_e(e^{j\omega}) A_o(e^{j\omega}) e^{-j\omega M/2} e^{-j\omega/2 + j\pi/2} \\ &= A'_o(e^{j\omega}) e^{j\omega M'/2 + j\pi/2} \end{aligned}$$

where $A'_o(e^{j\omega})$ is a real, odd, periodic function of ω and M' is an even integer.

Thus, the resulting system $H(e^{j\omega})$ has the form of a type III FIR generalized linear phase system. It is antisymmetric, has odd length (M is even), and has generalized linear phase.

Question 8

To find the input to a linear shift-invariant filter that will produce a given output $y(n)$, we use the relationship $Y(z) = H(z)X(z)$ to solve for $X(z)$:

$$X(z) = \frac{Y(z)}{H(z)}$$

Computing the z -transform of $y(n)$, we have

$$Y(z) = \frac{-\frac{1}{3}}{1 - \frac{1}{4}z^{-1}} + \frac{\frac{4}{3}}{1 - 2z^{-1}} = \frac{1 + \frac{1}{3}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - 2z^{-1})}$$

Therefore,

$$X(z) = \frac{(1 + \frac{1}{3}z^{-1})(1 - \frac{1}{2}z^{-1})}{(1 - \frac{1}{4}z^{-1})(1 - 2z^{-1})(1 + z^{-1})} = \frac{A}{1 - \frac{1}{4}z^{-1}} + \frac{B}{1 - 2z^{-1}} + \frac{C}{1 + z^{-1}}$$

where

$$A = \left(1 - \frac{1}{4}z^{-1}\right)X(z) \Big|_{z^{-1}=4} = \frac{1 - \frac{1}{6}z^{-1} - \frac{1}{6}z^{-2}}{(1 - 2z^{-1})(1 + z^{-1})} \Big|_{z^{-1}=4} = \frac{1}{15}$$

$$B = (1 - 2z^{-1})X(z) \Big|_{z^{-1}=\frac{1}{2}} = \frac{1 - \frac{1}{6}z^{-1} - \frac{1}{6}z^{-2}}{\left(1 - \frac{1}{4}z^{-1}\right)(1 + z^{-1})} \Big|_{z^{-1}=\frac{1}{2}} = \frac{2}{3}$$

$$C = (1 + z^{-1})X(z) \Big|_{z=-1} = \frac{1 - \frac{1}{6}z^{-1} - \frac{1}{6}z^{-2}}{\left(1 - \frac{1}{4}z^{-1}\right)(1 - 2z^{-1})} \Big|_{z=-1} = \frac{4}{15}$$

Because $h(n)$ is causal, the region of convergence for $H(z)$ is $|z| > \frac{1}{2}$. With the region of convergence of $Y(z)$ the annulus $\frac{1}{4} < |z| < 2$, the region of convergence of $X(z)$ is $\frac{1}{4} < |z| < 1$. Therefore,

$$x(n) = \frac{1}{15} \left(\frac{1}{4}\right)^n u(n) - \frac{2}{3} 2^n u(-n-1) - \frac{4}{15} (-1)^n u(-n-1)$$

Question 9

Because $Y(z)$ is related to $X(z)$ as follows:

$$Y(z) = \left(1 + \frac{1}{2}z^{-n_d} + \frac{1}{4}z^{-2n_d}\right)X(z)$$

the inverse filter is

$$G(z) = \frac{1}{1 + \frac{1}{2}z^{-n_d} + \frac{1}{4}z^{-2n_d}}$$

We must check, however, to see whether or not this filter is realizable. First, note that we may write $G(z)$ as

$$G(z) = F(z^{n_d})$$

where

$$F(z) = \frac{1}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

The poles of $F(z)$ are at

$$z = -\frac{1}{4}(1 \pm j\sqrt{3})$$

which are *inside* the unit circle at a radius of $r = 0.5$. Therefore, the poles of $G(z)$ are inside the unit circle, at a radius of $r' = (0.5)^{-n_d}$, and $G(z)$ is realizable.

Question 10

Multiplying a sequence by α^n moves the poles and zeros radially by a factor of α :

$$Y(z) = X\left(\frac{z}{\alpha}\right) = \frac{\left(1 - \frac{3}{2}\alpha z^{-1}\right)\left(1 + \frac{1}{3}\alpha z^{-1}\right)\left(1 + \frac{5}{3}\alpha z^{-1}\right)}{(1 - \alpha z^{-1})^2\left(1 - \frac{1}{4}\alpha z^{-1}\right)}$$

In order for $Y(z)$ to be minimum phase, all of the poles and zeros must be inside the unit circle. Because the singularity (pole or zero) of $X(z)$ that is the furthest from the unit circle is the zero at $z = -\frac{5}{3}$, $y(n)$ will be minimum phase if $|\alpha| < \frac{3}{5}$.

Question 11

This filter requires a maximal passband error of $\delta_p = 0.05$, and a maximal stopband error of $\delta_s = 0.1$. Converting these values to dB gives

$$\delta_p = -26 \text{ dB}$$

$$\delta_s = -20 \text{ dB}$$

This requires a window with a peak approximation error less than -26 dB. Looking in Table 7.1, the Hanning, Hamming, and Blackman windows meet this criterion.

Next, the minimum length L required for each of these filters can be found using the "approximate width of mainlobe" column in the table since the mainlobe width is about equal to the transition width. Note that the actual length of the filter is $L = M + 1$.

Hanning:

$$\begin{aligned} 0.1\pi &= \frac{8\pi}{M} \\ M &= 80 \end{aligned}$$

Hamming:

$$\begin{aligned} 0.1\pi &= \frac{8\pi}{M} \\ M &= 80 \end{aligned}$$

Blackman:

$$\begin{aligned} 0.1\pi &= \frac{12\pi}{M} \\ M &= 120 \end{aligned}$$

Question 12

- (a) It is well known that convolving two rectangular windows results in a triangular window. Specifically, to get the $(M+1)$ point Bartlett window for M even, we can convolve the following rectangular windows.

$$\begin{aligned} r_1[n] &= \begin{cases} \sqrt{\frac{2}{M}}, & n = 0, \dots, \frac{M}{2} - 1 \\ 0, & \text{otherwise} \end{cases} \\ r_2[n] &= r_1[n-1] \end{aligned}$$

Using the known transform of a rectangular window we have

$$\begin{aligned} W_{R_1}(e^{j\omega}) &= \sqrt{\frac{2}{M}} \frac{\sin(\omega M/4)}{\sin(\omega/2)} e^{-j\omega(\frac{M}{4} - \frac{1}{2})} \\ W_{R_2}(e^{j\omega}) &= \sqrt{\frac{2}{M}} \frac{\sin(\omega M/4)}{\sin(\omega/2)} e^{-j\omega(\frac{M}{4} + \frac{1}{2})} \\ W_B(e^{j\omega}) &= W_{R_1}(e^{j\omega}) W_{R_2}(e^{j\omega}) \\ &= \frac{2}{M} \left(\frac{\sin(\omega M/4)}{\sin(\omega/2)} \right)^2 e^{-j\omega M/2} \end{aligned}$$

Note: The Bartlett window as defined in the text is zero at $n = 0$ and $n = M$. These points are included in the $M + 1$ points.

For M odd, the Bartlett window is the convolution of

$$\begin{aligned} r_3[n] &= \begin{cases} \sqrt{\frac{2}{M}}, & n = 0, \dots, \frac{M-1}{2} \\ 0, & \text{otherwise} \end{cases} \\ r_4[n] &= \begin{cases} \sqrt{\frac{2}{M}}, & n = 1, \dots, \frac{M-1}{2} \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

In the frequency domain we have

$$\begin{aligned} W_{R_3}(e^{j\omega}) &= \sqrt{\frac{2}{M}} \frac{\sin(\omega(M+1)/4)}{\sin(\omega/2)} e^{-j\omega(\frac{M-1}{4})} \\ W_{R_4}(e^{j\omega}) &= \sqrt{\frac{2}{M}} \frac{\sin(\omega(M-1)/4)}{\sin(\omega/2)} e^{-j\omega(\frac{M-3}{4} + 1)} \\ W_B(e^{j\omega}) &= W_{R_3}(e^{j\omega}) W_{R_4}(e^{j\omega}) \\ &= \frac{2}{M} \left(\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} \right) \left(\frac{\sin[\omega(M-1)/2]}{\sin(\omega/2)} \right) e^{-j\omega M/2} \end{aligned}$$

(b)

$$w[n] = \left[A + B \cos\left(\frac{2\pi n}{M}\right) + C \cos\left(\frac{4\pi n}{M}\right) \right] w_R[n]$$

$$W(e^{j\omega}) = \left\{ 2\pi A \delta(\omega) + B\pi \left[\delta\left(\omega + \frac{2\pi}{M}\right) + \delta\left(\omega - \frac{2\pi}{M}\right) \right] + C\pi \left[\delta\left(\omega + \frac{4\pi}{M}\right) + \delta\left(\omega - \frac{4\pi}{M}\right) \right] \right\} \\ \otimes \left\{ \frac{\sin(\omega(M+1)/2)}{\sin(\omega/2)} e^{-j\omega M/2} \right\}$$

where $\otimes$ denotes periodic convolution.

(c) For the Hanning window $A = 0.5$, $B = -0.5$, and $C = 0$.

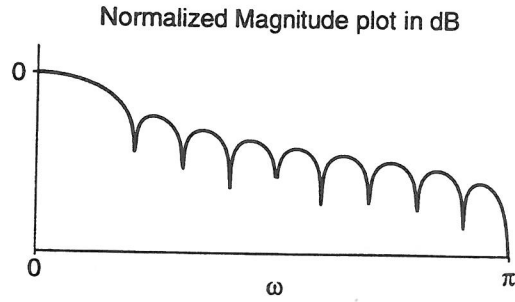
$$w_{\text{Hanning}}[n] = \left[0.5 - 0.5 \cos\left(\frac{2\pi n}{M}\right) \right] w_R[n]$$

$$W_{\text{Hanning}}(e^{j\omega}) = 0.5W_R(e^{j\omega}) - 0.25W_R(e^{j\omega}) \otimes \left[\delta\left(\omega + \frac{2\pi}{M}\right) + \delta\left(\omega - \frac{2\pi}{M}\right) \right] \\ = 0.5W_R(e^{j\omega}) - 0.25 \left[W_R(e^{j(\omega + \frac{2\pi}{M})}) + W_R(e^{j(\omega - \frac{2\pi}{M})}) \right]$$

where

$$W_R(e^{j\omega}) = \frac{\sin(\omega(M+1)/2)}{\sin(\omega/2)} e^{-j\omega M/2}$$

Below is a normalized sketch of the magnitude response in dB.



Question 13

For a type II linear phase filter,

$$h(n) = h(N - n)$$

where N is odd. Therefore, $h(n)$ is symmetric about the half-integer, $N/2$, and the symmetry condition is equivalent to

$$h\left(\frac{N+1}{2} - k\right) = h\left(\frac{N+1}{2} + k\right) \quad k = 1, \dots, \frac{N+1}{2}$$

Thus, the frequency response may be written as

$$H(e^{j\omega}) = \sum_{n=0}^N h(n)e^{-jn\omega} = \sum_{k=1}^{(N+1)/2} h\left(\frac{N+1}{2} - k\right)e^{-j(\frac{N+1}{2}-k)\omega} + \sum_{k=1}^{(N+1)/2} h\left(\frac{N+1}{2} + k\right)e^{-j(\frac{N+1}{2}+k)\omega}$$

Factoring out a linear phase term $e^{-jN\omega/2}$ and using the symmetry of $h(n)$, we have

$$\begin{aligned} H(e^{j\omega}) &= e^{-jN\omega/2} \sum_{k=1}^{(N+1)/2} h\left(\frac{N+1}{2} - k\right) \left[e^{j(k-\frac{1}{2})\omega} + e^{-j(k-\frac{1}{2})\omega} \right] \\ &= e^{-jN\omega/2} \sum_{k=1}^{(N+1)/2} 2h\left(\frac{N+1}{2} - k\right) \cos\left[\left(k - \frac{1}{2}\right)\omega\right] \end{aligned} \quad (5.21)$$

Therefore,

$$H(e^{j\omega}) = e^{-jN\omega/2} \sum_{k=1}^{(N+1)/2} b(k) \cos\left[\left(k - \frac{1}{2}\right)\omega\right]$$

where

$$b(k) = 2h\left(\frac{N+1}{2} - k\right) \quad k = 1, 2, \dots, \frac{N+1}{2}$$

which is the desired result.